MPG No. 2.1.3 $\bar{1}$ [Type I, triclinic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & M_y & M_z \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$
$$M_y = \mathbb{M}_y^{(1)}$$
$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & M_{xxy} & M_{xxz} \\ M_{yyx} & M_{yyy} & M_{yyz} \\ M_{zzx} & M_{zzy} & M_{zzz} \\ M_{yzx} & M_{yyz} & M_{zzy} \\ M_{xxz} & M_{yzx} & M_{zzx} \\ M_{xxy} & M_{yyx} & M_{yzx} \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$
$$M_{xxy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\beta(1)}$$
$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\beta(1)}$$
$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$
$$M_{yyy} = 3\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)}$$
$$M_{yyz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\beta(1)}$$
$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$
$$M_{zzy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\beta(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & M_{zzy} & 0 \\ M_{yzx} & -\frac{M_{yyz}}{2} & -\frac{M_{zzy}}{2} \\ -\frac{M_{xxz}}{2} & M_{zxy} & -\frac{M_{zzx}}{2} \\ -\frac{M_{xxy}}{2} & -\frac{M_{yyx}}{2} & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{xxy} = 2\mathbb{M}_{y}^{(2)} - 2\mathbb{T}_{xz}^{(1)}$$
$$M_{xxz} = 2\mathbb{M}_{z}^{(2)} + 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yyx} = 2\mathbb{M}_{x}^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$
$$M_{yyz} = 2\mathbb{M}_{z}^{(2)} - 2\mathbb{T}_{xy}^{(1)}$$
$$M_{zzx} = 2\mathbb{M}_{x}^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$
$$M_{zzy} = 2\mathbb{M}_{y}^{(2)} + 2\mathbb{T}_{xz}^{(1)}$$
$$M_{yzx} = 3\mathbb{T}_{u}^{(1)} + \mathbb{T}_{v}^{(1)}$$
$$\frac{M_{yyz}}{2} = \mathbb{M}_{z}^{(2)} - \mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{zzy}}{2} = \mathbb{M}_{y}^{(2)} + \mathbb{T}_{xz}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_{z}^{(2)} + \mathbb{T}_{xy}^{(1)}$$
$$M_{zxy} = -3\mathbb{T}_{u}^{(1)} + \mathbb{T}_{v}^{(1)}$$
$$\frac{M_{zzx}}{2} = \mathbb{M}_{x}^{(2)} - \mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{xxy}}{2} = \mathbb{M}_{y}^{(2)} - \mathbb{T}_{xz}^{(1)}$$
$$\frac{M_{yyx}}{2} = \mathbb{M}_{x}^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & M_{yzz} \\ M_{zxx} & M_{zxy} & M_{zxz} \\ M_{xyx} & M_{xyy} & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{0}^{(1)} - \mathbb{T}_{u}^{(2)} + \mathbb{T}_{v}^{(2)}$$
$$M_{yzy} = \mathbb{M}_{z}^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{yzz} = -\mathbb{M}_{y}^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{zxx} = -\mathbb{M}_{z}^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{zxy} = \mathbb{T}_{0}^{(1)} - \mathbb{T}_{u}^{(2)} - \mathbb{T}_{v}^{(2)}$$
$$M_{zxz} = \mathbb{M}_{x}^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyx} = \mathbb{M}_{y}^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{xyy} = -\mathbb{M}_{x}^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyz} = \mathbb{T}_{0}^{(1)} + 2\mathbb{T}_{u}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).